Name: Thamaraiselvi S

Reg No: 192511042

Dept: B.E CSE

CSA 0908 – Programming in Java

Slot - B

SectionA:Multiple-Choice

Questions (10Questions)

1. Mixed Types and Integer Division

```
int a = 7, b = 2;
double r = a / b + a % b * 0.5;
System.out.println(r);
```

A) 3.0

B) 4.0

C) 3.5

D) 4.5

correct answer is C) 3.5.

2. Identify the Error

```
byte x = 10;
byte y = 20;
byte z = x + y;
System.out.println(z);
```

A) 30

B) Compilation error

C) -26

D) Runtime exception

Correct answer: B) Compilation error

3. Unary Operators and Precedence

```
int a = 4, b = 6;
int c = a++ * 2 + --b;
System.out.println(a + " " + b + " " + c);
```

A) 5 6 14

B) 4 5 13

C) 5 5 14

D) 5 5 13

correct answer is D) 5 5 13.

4. Which if Does the else Belong To?

```
int x = 5;
if (x > 3)
 if (x > 10)
 System.out.println("A");
else
 System.out.println("B");
```

A) B

B) A

C) Both A and B are printed
D) Nothing is printed

Correct answer: A) B

5. Switch and Missing break

```
int day = 2;
switch (day) {
 case 1:
System.out.print("Mon
");  case 2:
System.out.print("Tue
");
 case 3:
System.out.print("Wed ");
break;  case 4:
System.out.print("Thu ");
default:
System.out.print("Off");
}
```

A) Tue

B) Mon Tue Wed

C) Tue Wed

D) Tue Wed Thu Off

Correct answer: C) Tue Wed

6. Switch Rules

```
double grade = 2.0;
switch (grade) {
 case 2.0:
System.out.println("Two");
break; default:
System.out.println("Other"); }
```

A) Compilation error

B) Two

C) Other

D) Nothing is printed

Correct answer: A) Compilation error

7. Short-Circuit Evaluation

```
int a = 5, b = 10;
boolean r = (a > 5) && (++b >
10) || (b++ > 9);
System.out.println(a + " " + b +
" " + r);
```

A) 5 10 false

B) 5 11 false

C) 5 12 true

D) 5 11 true

Correct answer: D) 5 11 true

8. char Arithmetic and Type Promotion

```
char c = 'A';
c += 2;
System.out.println(c + 1);
```

A) D

B) 68

C) 67

D) Compilation error

Correct answer: B) 68

9. Ternary Operator with Side Effects

```
int a = 8, b = 12;
int m = (a > b) ? a++ : b--;
System.out.println(a + " " + b + " " + m);
```

A) 8 11 12

B) 9 12 8

C) 8 11 11

D) 9 11 12

Correct answer: A) 8 11 12

10. Input and Output Methods

The user types `12 5` (separated by a space) when the program runs.

```
Scanner sc = new Scanner(System.in);
int a = sc.nextInt();
int b = sc.nextInt();
System.out.print(a + b);
```

System.out.println(a - b);
A) 17 7

B) 24

C) 17 and 7 on two separate lines

D) 177

Correct answer: D) 177

Section B: Scenario-Based Programming Questions (3 Questions)

Problem 1: Candy Sharing

Scenario: A teacher wants to share candies equally among the students in her class. Any candies that cannot be shared equally will be kept by her.

Problem Statement: Write a program that reads the total number of candies and the number of students. Display how many candies each student gets and how many candies are left over.

Input Format: Two integers on one line: the number of candies,

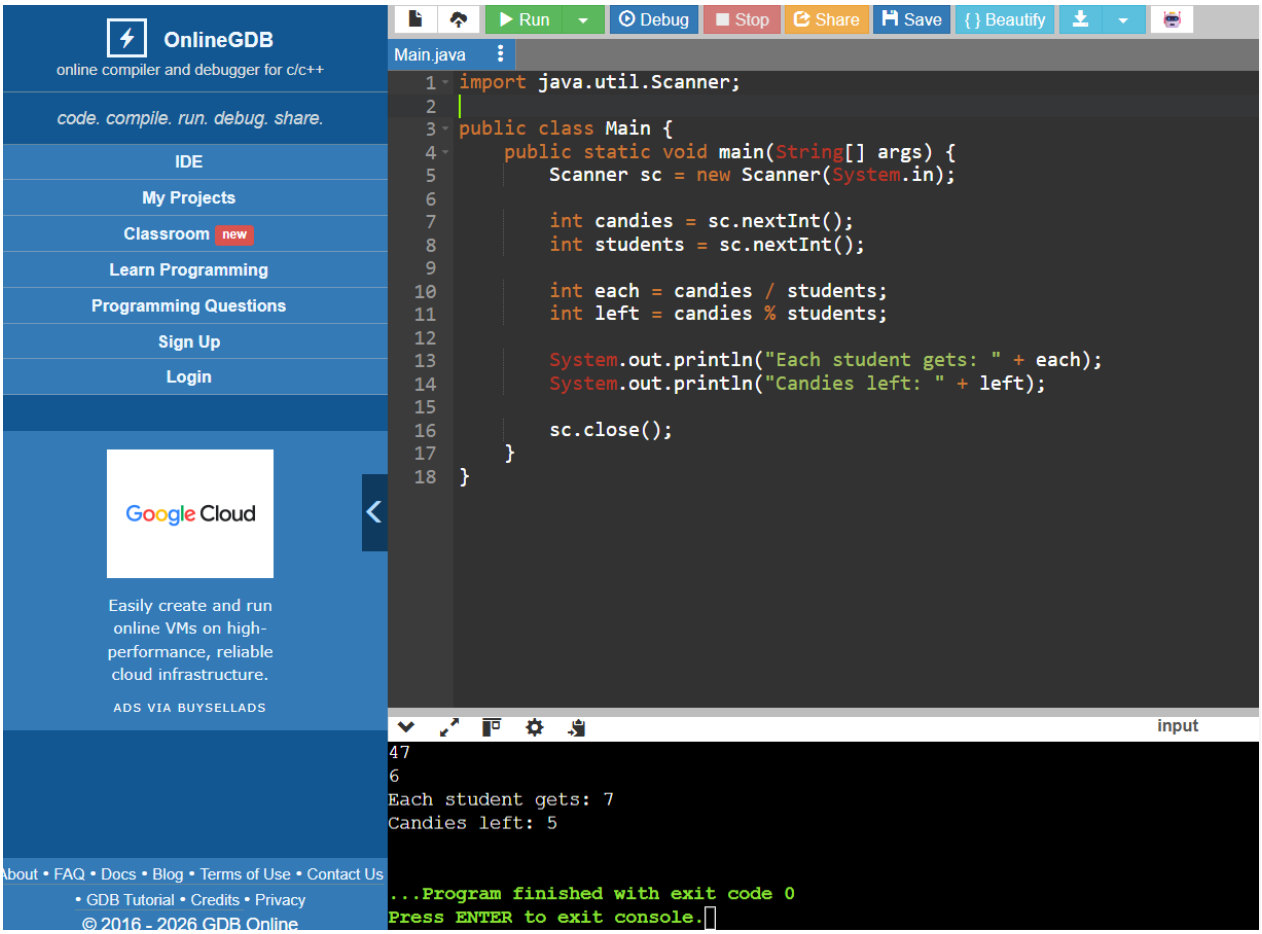
then the number of students. **Output Format:** Two lines:

Each student gets: <number>
Candies left: <number>
Constraints:

- $0 \leq candies \leq 10000$
- $1 \leq students \leq 100$

Sample Input:
47 6
Sample Output:

Each student gets: 7
Candies left: 5



Problem 2: Amusement Park Ride Eligibility

Scenario: An amusement park has a roller coaster with safety rules. A rider is allowed on the ride if **either** of these is true:

- The rider is at least 10 years old **and** at least 120 cm tall; **or**
- The rider is at least 100 cm tall **and** is accompanied by an adult.

Problem Statement: Write a program that reads the rider's age, height, and whether an adult accompanies them, and decides if the rider is allowed.

Input Format: Three integers on one line: age (in years), height (in cm), and adult accompanying (1 for yes, 0 for no).

Output Format: Print ALLOWED or NOT ALLOWED.

Constraints:

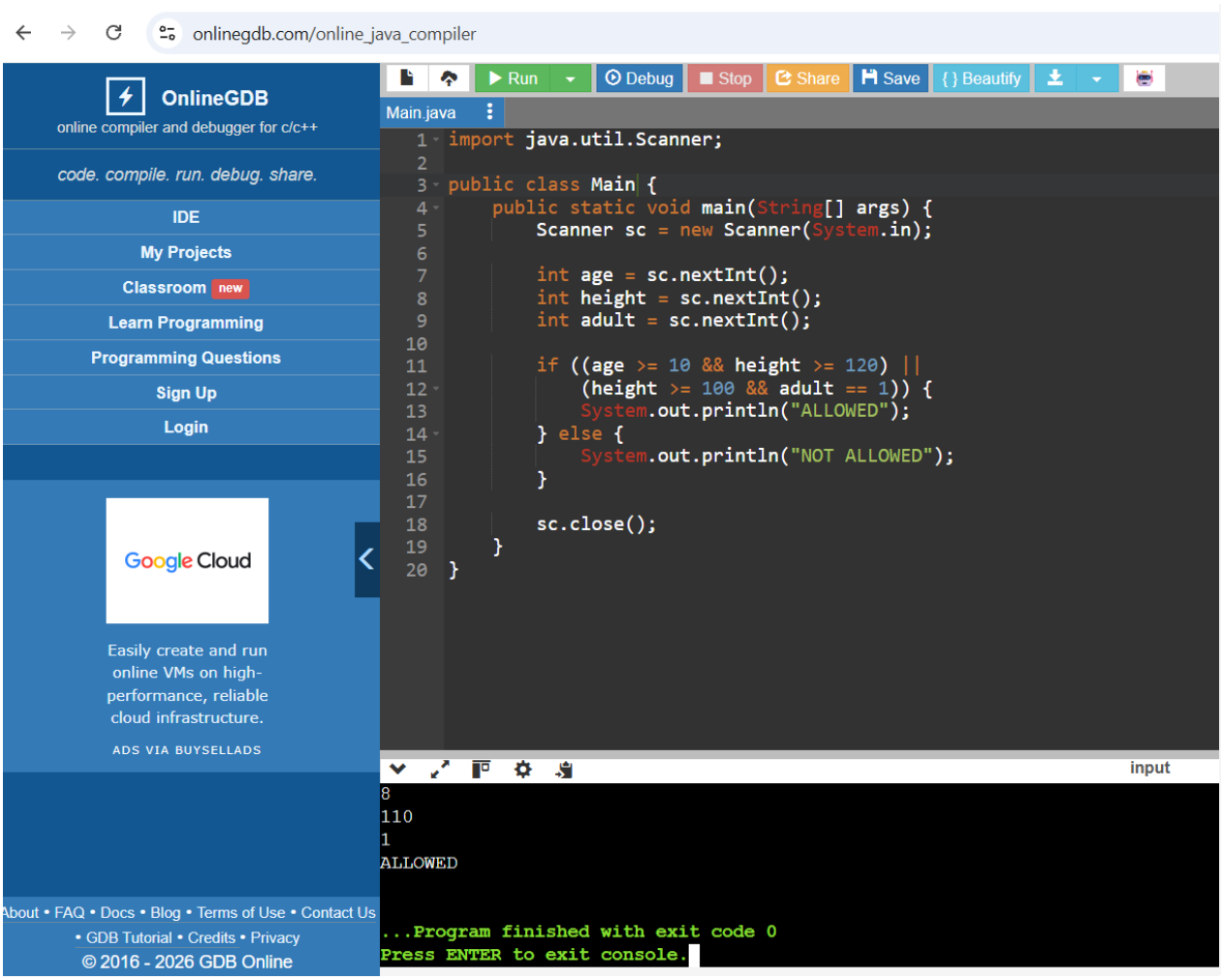
- $1 \leq age \leq 90$
- $50 \leq height \leq 220$
- The third value is either 0 or 1

Sample Input 1:

```
8 110 1
```

Sample Output 1:

```
ALLOWED
```

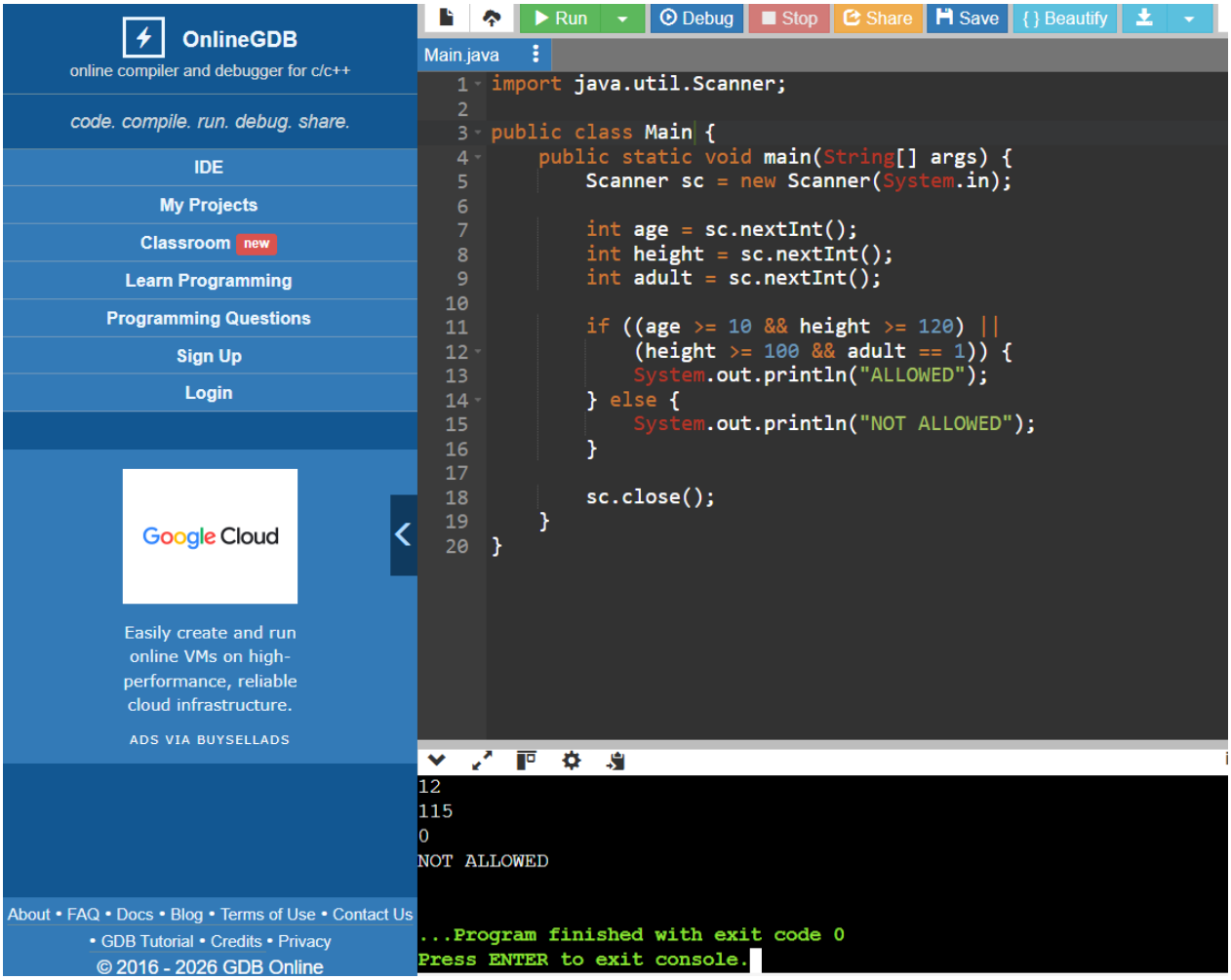


Sample Input 2:

12 115 0

Sample Output 2:

NOT ALLOWED


OnlineGDB
online compiler and debugger for c/c++
code. compile. run. debug. share.
IDE
My Projects
Classroom new
Learn Programming
Programming Questions
Sign Up
Login
Google Cloud
Easily create and run online VMs on high-performance, reliable cloud infrastructure.
ADS VIA BUYSELLADS
About • FAQ • Docs • Blog • Terms of Use • Contact Us
• GDB Tutorial • Credits • Privacy
© 2016 - 2026 GDB Online
Run
Debug
Stop
Share
Save
{ } Beautify
Main.java
1 import java.util.Scanner;
2
3 public class Main {
4 public static void main(String[] args) {
5 Scanner sc = new Scanner(System.in);
6
7 int age = sc.nextInt();
8 int height = sc.nextInt();
9 int adult = sc.nextInt();
10
11 if ((age >= 10 && height >= 120) ||
12 (height >= 100 && adult == 1)) {
13 System.out.println("ALLOWED");
14 } else {
15 System.out.println("NOT ALLOWED");
16 }
17
18 sc.close();
19 }
20 }
12
115
0
NOT ALLOWED
...Program finished with exit code 0
Press ENTER to exit console.

Problem 3: College Canteen Billing

Scenario: The college canteen uses a numbered menu:

Code	Item	Price (Rs.)
1	Veg Burger	60
2	Sandwich	50
3	Pasta	120
4	Cold Coffee	80

A customer orders one type of item in some quantity. If the bill is **Rs. 500 or more**, a **10% discount** is given on the bill.

Problem Statement: Write a program that reads the item code and quantity and prints the final amount payable. If the item code is not between 1 and 4, print an error message instead.

Input Format: Two integers on one line: item code, then quantity.

Output Format:

- For a valid code: `Amount payable: <amount>` (the amount as a decimal number, for example `540.0`)
- For an invalid code: `Invalid choice`

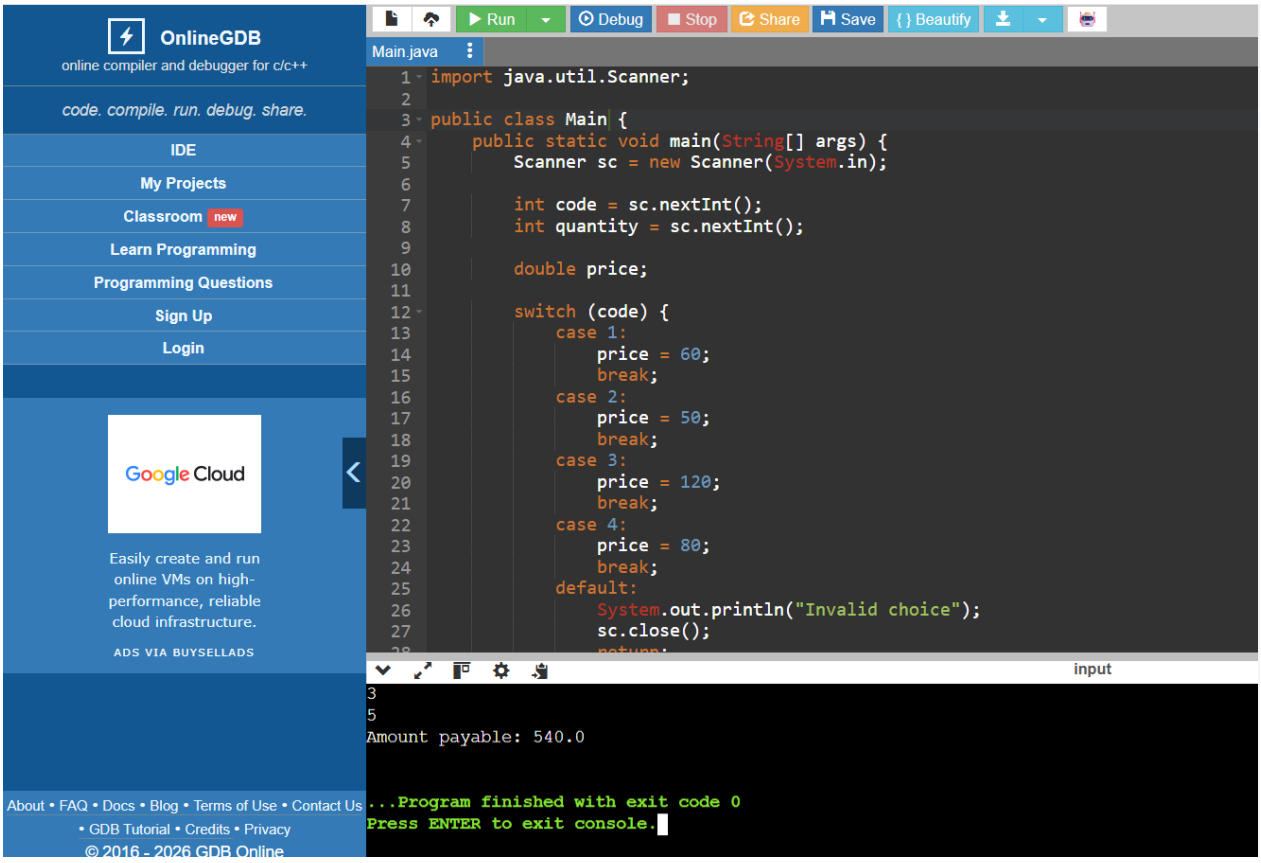
Constraints: `1 ≤ quantity ≤ 50`

Sample Input 1:

```
3 5
```

Sample Output 1:

```
Amount payable: 540.0
```

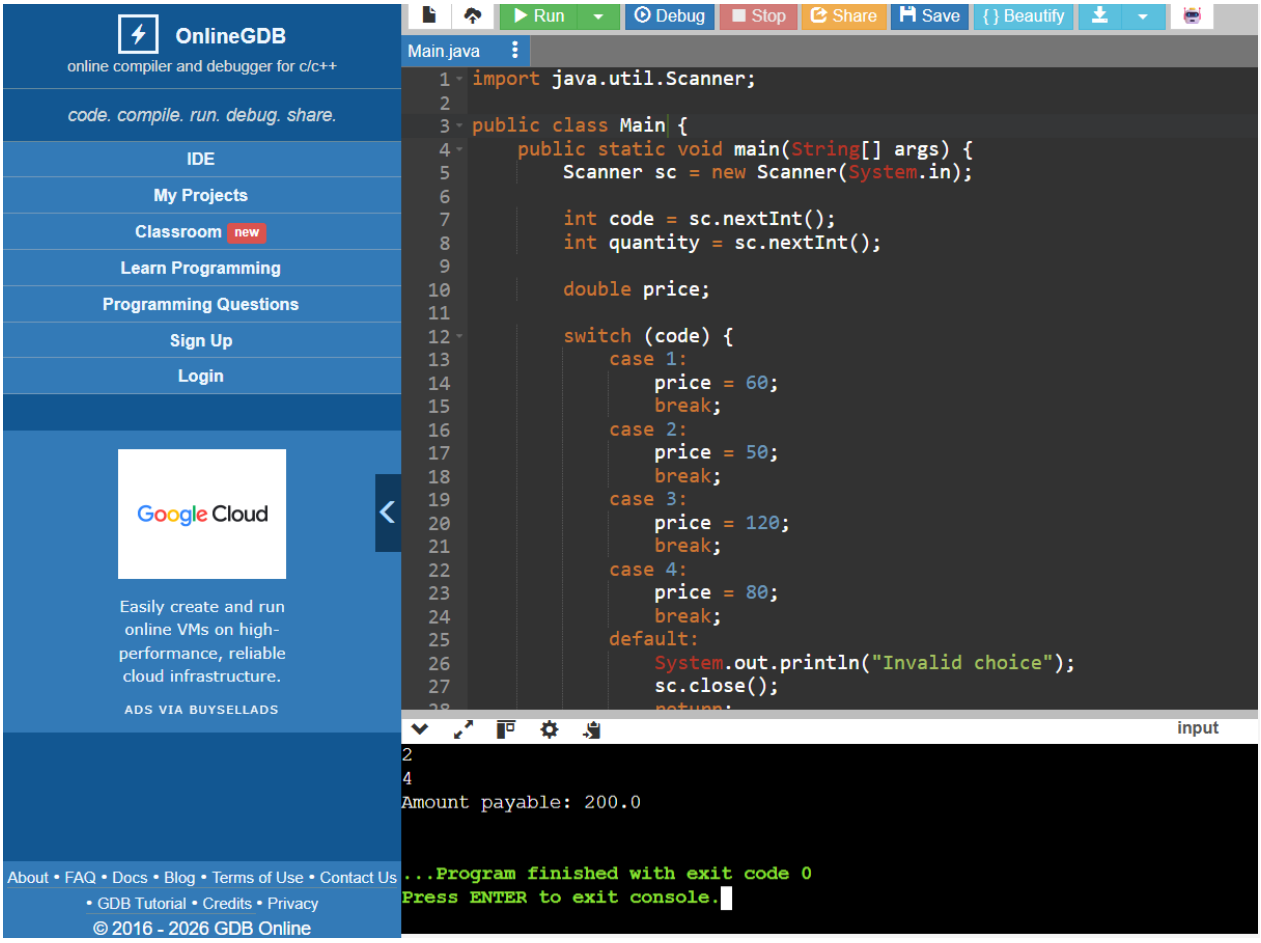


Sample Input 2:

2 4

Sample Output 2:

Amount payable: 200.0

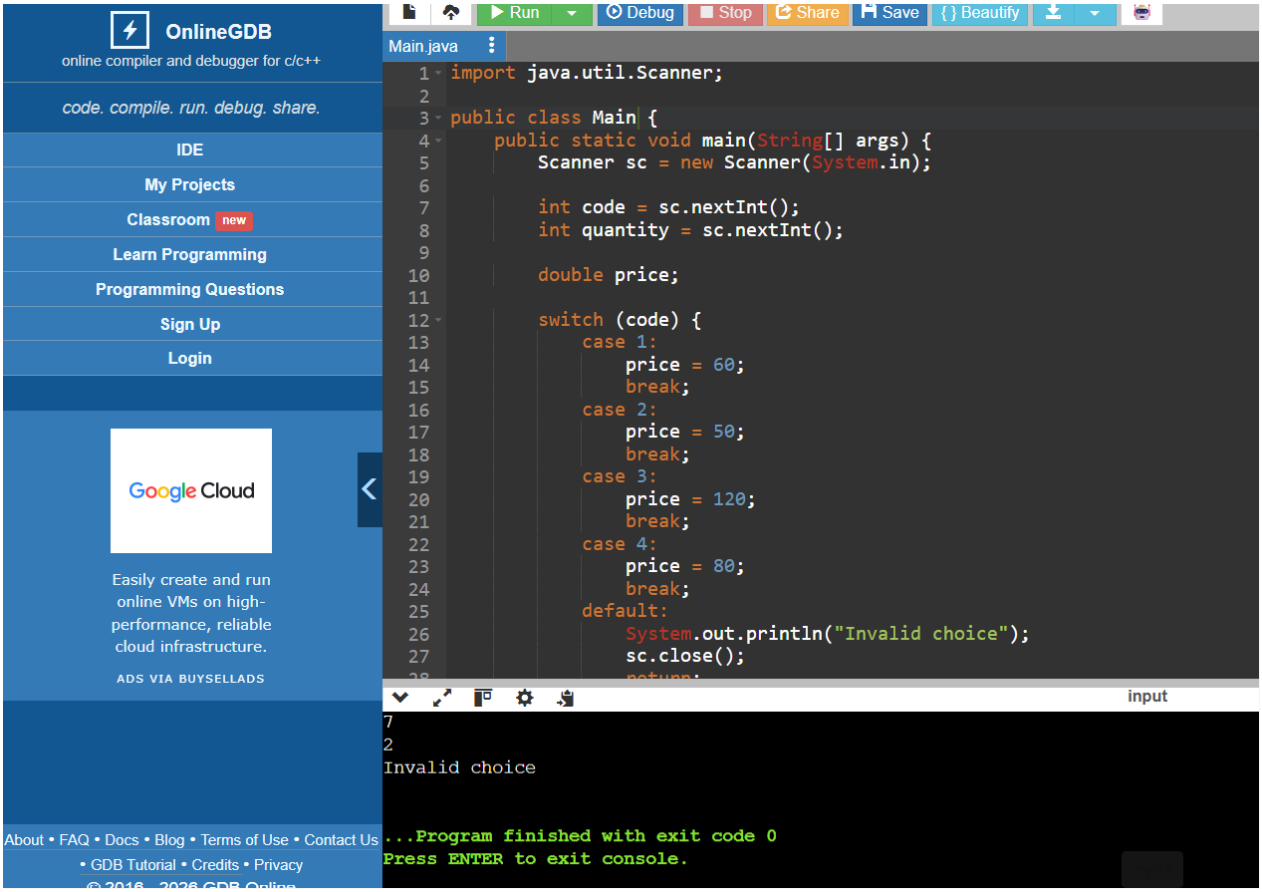


Sample Input 3:

7 2

Sample Output 3:

Invalid choice


OnlineGDB
online compiler and debugger for c/c++
code. compile. run. debug. share.
IDE
My Projects
Classroom new
Learn Programming
Programming Questions
Sign Up
Login
Google Cloud
Easily create and run online VMs on high-performance, reliable cloud infrastructure.
ADS VIA BUYSELLADS
About • FAQ • Docs • Blog • Terms of Use • Contact Us
• GDB Tutorial • Credits • Privacy
Run
Debug
Stop
Share
Save
{ } Beautify
Main.java
import java.util.Scanner;
public class Main {
public static void main(String[] args) {
Scanner sc = new Scanner(System.in);
int code = sc.nextInt();
int quantity = sc.nextInt();
double price;
switch (code) {
case 1:
price = 60;
break;
case 2:
price = 50;
break;
case 3:
price = 120;
break;
case 4:
price = 80;
break;
default:
System.out.println("Invalid choice");
sc.close();
input
7
2
Invalid choice
...Program finished with exit code 0
Press ENTER to exit console.